VBCores Servo Board v0.2

Motor supply voltage selection jumper
Open: 12 V.
Closed: 24 V.
This setting applies only when the motor supply source is DC-DC.
The motor supply voltage cannot exceed VIN.

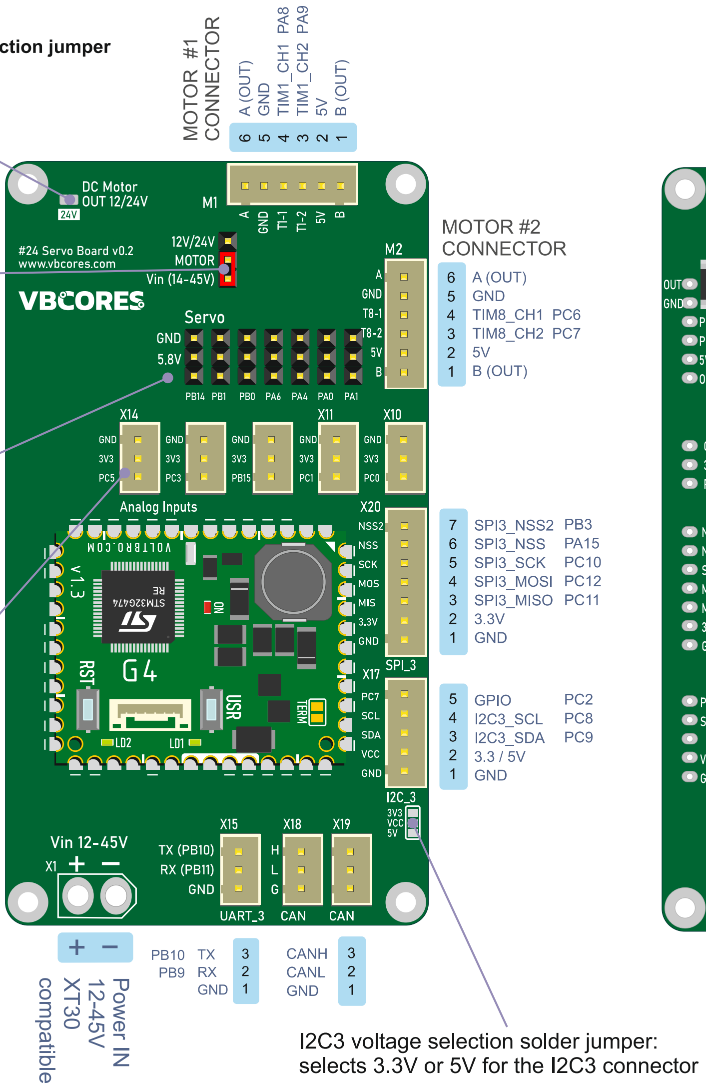
Motor supply source selection jumper.
1. DC-DC output: 12V or 24V, 5 A max.
2. Raw VIN: 12-45V, unregulated.
The motor supply voltage cannot exceed VIN.

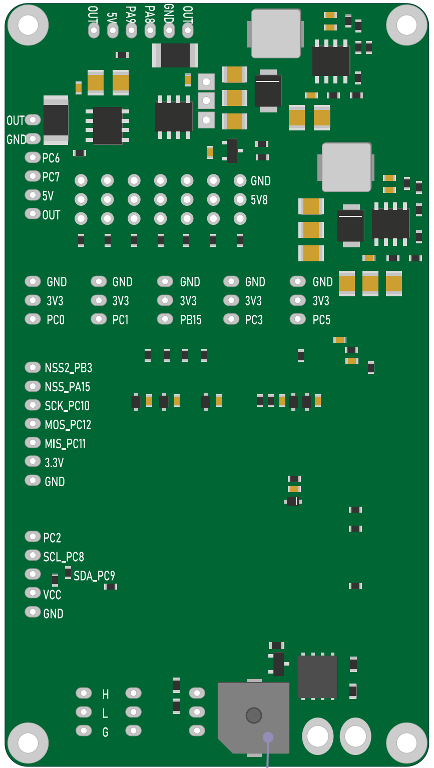
SERVO, 7ch
JST connector

GND | 5.8V, 5A max | PB14, PB1, PB0, PA6, PA4, PA0, PA1

Analog in 5ch, **3.3V MAX**
XH2.54 connector

GND | 3.3V | PC5, PC3, PB15, PC1, PC0





Buzzer, PA11

Motor driver

TI DRV8870, 45V, 3.5A
Brushed DC Motor Driver (PWM Control)

Control	PIN	Notes
IN1 (Motor 1)	PB6	TIM4_CH1
IN2 (Motor 1)	PB4	
IN1 (Motor 2)	PB7	TIM4_CH2
IN2 (Motor 2)	PB5	
VREF	PB2	3.3A max
R_ISEN		0,15 Ohm

Input voltage sensing

Divider ratio: 1:16

Control	PIN	ADC
VIN_SENSE	PA7	ADC2_IN4

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VIN: 12-45 V DC
MCU: VB32G4 (STM32G474RE)
7 × servo outputs, 5.8V, 5A max
5 × analog/GPIO inputs, 3.3V max
2 × DRV8870 brushed DC motor drivers
Motor supply: 12/24 V DC-DC output, 5A max, or raw VIN
2 × A/B encoder interfaces
Interfaces: SPI, I2C, UART, CAN/CAN FD
Input voltage monitoring
Buzzer
Dimensions: 56 × 101 mm
Mounting holes: 50 × 95 mm, D2.5 mm

NOTES:

1. Motor drivers start operating at a minimum supply voltage of 9 V.
2. Maximum motor current is limited to 3.3 A.
3. For motors without encoders, use only the Motor A and Motor B terminals.
4. Servo supply voltage is 5.8 V. Make sure your servos support this voltage.
5. Servo signal pins use 3.3 V logic. Do not apply voltages above 3.3 V.
6. Maximum voltage on analog input pins is 3.3 V.
7. Maximum voltage on the SPI3 connector is 3.3 V.
8. I2C3 lines are compatible with both 3.3 V and 5 V logic levels.
9. UART logic level is 3.3 V.

VBCores
www.vbcores.com
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